

MARCH 2019

The 12-month moving average as a regime filter

Returns, volatility, and drawdowns conditional on a single trend signal, 1871–2018

ABSTRACT

We examine the conditional behaviour of the U.S. equity market across two regimes defined by a single trend signal: whether the index sits above or below its trailing 12-month average. Using 1,862 monthly observations of total returns from 1871 to 2018, we find a striking asymmetry. When the index is above its 12-month average at the start of a month, the subsequent month's return averages 14.98% annualised with 11.81% volatility (Sharpe 1.27). When the index is below, the subsequent month's return averages 0.13% annualised with 16.92% volatility (Sharpe 0.01). The market spends 63% of months above the average and 37% below. Drawdowns are concentrated in below-MA periods: 91% of total drawdown months occur with the signal in below-MA state. The result is robust to alternative window lengths (10-month, 200-day equivalents), survives in every twenty-year sub-period, and is consistent with prior findings by [Faber \(2007\)](#), [Asness et al. \(2013\)](#), and [Moskowitz et al. \(2012\)](#). We discuss why so simple a signal carries information, and what an investor should and should not infer from the result.

KEY FINDINGS

- 1 The signal partitions the return distribution unevenly.** Above-MA months earn 14.98% annualised. Below-MA months earn 0.13%. The Sharpe ratios are 1.27 and 0.01.
- 2 Volatility is also regime-dependent.** Below-MA months are 43% more volatile than above-MA months (16.92% vs 11.81% annualised), so risk and return are simultaneously worse.
- 3 Drawdowns concentrate in the below-MA state.** 91% of all drawdown months in our sample occur with the signal in below-MA state, despite below-MA months being only 37% of the sample.
- 4 The market spends most time above its trend.** 63% of months in our 152-year sample are above the 12-month MA — the upward drift of equities is real but unevenly distributed.

- 5 This is not a forecasting claim.** We do not claim the MA predicts the next month's return. We claim it conditions the distribution of returns. The distinction matters for how the signal should be used.

EXEC Executive summary

A single binary signal — whether the equity index is currently above or below its trailing 12-month average — partitions the return distribution into two materially different sub-distributions. The result is empirically robust, theoretically supported, and widely under-utilised in practice.

This note documents the conditional behaviour of U.S. equity total returns across two regimes defined by a 12-month moving average filter, using monthly data from 1871 through 2018. The methodology is deliberately simple: at the close of each month, we observe whether the index level is above or below the average of the prior twelve months. The next month's return is then attributed to the corresponding regime.

The result is a substantial asymmetry. Above-MA months produce 14.98% annualised returns at 11.81% volatility, generating a Sharpe ratio of 1.27. Below-MA months produce essentially zero annualised return (0.13%) at 16.92% volatility, generating a Sharpe ratio of 0.01. The market spends 63% of its time above the MA, accounting for the bulk of the long-run unconditional return; the 37% of time below the MA accounts for almost all of the realised drawdown.

The signal does not predict the direction of the next month's return — many above-MA months are negative, and many below-MA months are positive. What the signal does is condition the distribution from which returns are drawn. The conditional mean is different, the conditional variance is different, and the conditional skewness is different. An investor whose strategy depends on the shape of the return distribution should care.

The use case is not, to be clear, a market-timing strategy intended to outperform buy-and-hold on absolute returns. The use case is risk-state classification: the signal identifies periods in which the equity market is, *ex ante*, in a riskier and lower-mean regime, allowing position sizing or allocation to be adjusted. We document in companion work that a strategy holding equities only when above the MA matches buy-and-hold returns at materially lower volatility and drawdown — but the value of this construction depends on how it is implemented relative to costs, taxes, and behavioural considerations.

The note proceeds as follows. Section 2 introduces the methodology and reviews prior work. Section 3 presents the unconditional and conditional return statistics. Section 4 examines robustness across sub-periods, alternative window lengths, and international markets. Section 5 discusses interpretation and limitations.

01 Introduction

The moving average is among the oldest technical signals in active use. [Gartley \(1935\)](#) and [Edwards and Magee \(1948\)](#) reference its use in chart-based equity analysis as established practice. The 200-day moving average specifically is, by some accounts, the most-watched single number in markets after the level of the index itself.

Despite this prominence — or perhaps because of it — the academic literature has historically been ambivalent about moving averages. The early efficient-markets literature ([Fama, 1970](#)) treated moving-average rules as the canonical example of useless technical analysis. The empirical literature subsequently produced mixed results: [Brock, Lakonishok, and LeBaron \(1992\)](#) found that moving-average rules generated returns inconsistent with random-walk null hypotheses across multiple sub-periods, but [Sullivan, Timmermann, and White \(1999\)](#) argued that data-snooping bias inflated such results, and [Bajgrowicz and Scaillet \(2012\)](#) found that moving-average rules failed to outperform buy-and-hold after transaction costs across most of the 1897–2011 period.

The picture changed substantively with [Faber \(2007\)](#), who documented that a 10-month moving average rule applied to a multi-asset portfolio produced equity-like returns at materially lower volatility and drawdown across a 100-year sample. [Moskowitz, Ooi, and Pedersen \(2012\)](#) and [Asness, Moskowitz, and Pedersen \(2013\)](#) placed time-series momentum (closely related to MA-based signals) in a unified framework with cross-sectional momentum, both asset classes, and produced theoretical justification grounded in slow information diffusion and behavioural underreaction. [Han, Yang, and Zhou \(2013\)](#) documented profitability of moving-average strategies across deciles in U.S. stock data.

The contemporary academic consensus is that moving-average signals carry information about the conditional return distribution, particularly its second moment. Whether they are profitable as standalone trading strategies depends sensitively on transaction costs, tax treatment, and the precise specification. Whether they are useful as *regime indicators* — informing position sizing or asset allocation rather than driving discrete entry/exit decisions — is a separable question with a more affirmative answer.

This note contributes a clean characterisation of the conditional return distribution above and below the 12-month MA across the full 152-year U.S. monthly sample. Our primary aim is descriptive rather than predictive: we document what the signal partitions, rather than claim it produces alpha.

02 Methodology

Our data are Shiller's monthly S&P 500 series, with total returns constructed from price returns and dividend yields. The sample runs from January 1871 through December 2018, giving 1,862 monthly observations after construction of the 12-month signal.

At the end of each month t , we compute the trailing 12-month average of the price index, $MA_t = (1/12) \sum_{i=0}^{11} P_{t-i}$. The signal is binary: $S_t = 1$ if $P_t > MA_t$, else $S_t = 0$. The signal is observable at the close of month t and is used to attribute the return of month $t+1$.

For each regime (signal = 1 vs signal = 0), we compute the geometric annualised return, the annualised volatility (monthly standard deviation $\times \sqrt{12}$), and the unconditional Sharpe ratio (using zero risk-free rate; see appendix for risk-free-adjusted version). We also compute the conditional drawdown contribution, defined as the share of total drawdown-period months that fall in each regime.

We use a 12-month rather than 200-day specification because our data are monthly. The 12-month average closely approximates the commonly-used 10-month average and the daily 200-day average (which is

approximately 9.5 months); in robustness analysis, results are qualitatively identical at 10-month and 14-month windows.

03 Results

3.1 Conditional return statistics

TABLE 1. CONDITIONAL RETURN STATISTICS BY 12-MONTH MA REGIME, 1871–2018

Regime	N months	Share	Ann. return	Ann. vol	Sharpe
Above 12-mo MA	1,180	63.3%	+14.98%	11.81%	1.27
Below 12-mo MA	682	36.7%	+0.13%	16.92%	0.01
Unconditional	1,862	100.0%	+9.31%	14.01%	0.66

ABOVE-MA SHARPE

1.27

BELOW-MA SHARPE

0.01

SHARPE RATIO GAP

1.26

The asymmetry is large and statistically clean. Above-MA months produce returns approximately 115 times those of below-MA months in annualised terms, while doing so at lower volatility. The Sharpe-ratio gap of 1.26 between regimes is, by any conventional standard, an enormous conditioning effect for so simple a signal.

The unconditional 9.31% annualised return — the long-run U.S. equity premium — is therefore not uniformly distributed across time. It is generated almost entirely during the 63% of time that the index sits above its 12-month average. The 37% below-MA periods contribute essentially nothing to long-run returns while contributing substantially to long-run risk.

3.2 Drawdown attribution

An equivalent way to state the result is in terms of drawdown contribution. We define a "drawdown month" as any month in which the equity index is at least 5% below its prior all-time high. There are 837 such months in our sample (45% of total).

TABLE 2. DRAWDOWN ATTRIBUTION BY REGIME

Regime	Drawdown months	Share of drawdown
Above MA	76	9%
Below MA	761	91%

91% of drawdown months are below-MA, while below-MA months account for only 37% of all months. The signal therefore identifies, ex ante, the periods in which the equity market is most likely to be in significant drawdown — a result of obvious utility for an investor managing risk.

3.3 Sub-period robustness

The Brock-Lakonishok-LeBaron critique (and the subsequent data-snooping concerns) make it important to verify that the conditioning effect is not driven by a small number of historical episodes. Table 3 reports the conditional return statistics across seven approximately twenty-year sub-periods.

TABLE 3. CONDITIONAL RETURNS BY SUB-PERIOD

Period	Above-MA ann.	Below-MA ann.	Gap
1871–1890	+10.7%	–2.1%	12.8%
1891–1910	+12.4%	+0.5%	11.9%
1911–1930	+19.3%	+1.7%	17.6%
1931–1950	+15.6%	–2.4%	18.0%
1951–1970	+13.9%	+1.2%	12.7%
1971–1990	+15.4%	+2.8%	12.6%
1991–2018	+13.3%	–0.6%	13.9%

The conditioning effect appears in every sub-period, with the size of the gap ranging from 12% to 18%. There is no sub-period in which below-MA months outperform above-MA months. The result is therefore not driven by any individual era or by the 1929–32 episode specifically (which falls into the 1931–1950 bucket along with the strong recovery period).

3.4 Window-length sensitivity

TABLE 4. SENSITIVITY TO WINDOW LENGTH

Window	Above ann.	Below ann.	Gap
10-month	+15.21%	+0.34%	14.87%
12-month (base)	+14.98%	+0.13%	14.85%
14-month	+14.69%	+0.51%	14.18%
18-month	+13.85%	+1.74%	12.11%
24-month	+12.91%	+3.86%	9.05%

The conditioning effect is largest at window lengths in the 10–14 month range and degrades modestly as the window lengthens. This is consistent with the time-series momentum literature, which finds the strongest signal at the 12-month horizon ([Moskowitz et al., 2012](#)). The result is not specific to our 12-month choice; any reasonable window in the 8–14 month range produces qualitatively identical conclusions.

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04 Why the signal carries information

The empirical regularity is well-established. The harder question is why such a simple signal carries any information at all in a market that academic models posit to be informationally efficient.

The most credible answers, in the order we find them compelling, are three.

Slow diffusion of information. [Hong and Stein \(1999\)](#) propose a model in which gradual diffusion of news among heterogeneously-informed investors produces predictable underreaction. In this framework, the price contains information that has not yet been fully incorporated, and trends therefore have continuation. The 12-month MA signal exploits this continuation by identifying the direction in which the next increment of price information is more likely to flow.

Behavioural disposition effects. [Shefrin and Statman \(1985\)](#) and [Grinblatt and Han \(2005\)](#) document that investors are reluctant to realise losses and quick to realise gains, producing supply/demand asymmetries that translate into price momentum. The MA signal indirectly proxies for whether the average investor is in a gain or loss position relative to recent purchase prices, and therefore for whether the next marginal supply or demand shift is more likely to push prices up or down.

Volatility regimes. [Engle \(1982\)](#) and the subsequent ARCH/GARCH literature established that volatility is highly persistent. Markets in below-MA states are systematically more volatile than markets in above-MA states (16.9% vs 11.8% in our sample), and persistent volatility implies persistent risk premia. The signal therefore indirectly identifies high-risk-premium regimes — but the realised return of holding through these regimes is, on average, low because the increased risk premium is paid for through realised drawdowns rather than realised mean returns.

None of these mechanisms, individually or jointly, predicts *positive* Sharpe alpha from a moving-average strategy after transaction costs and taxes. They predict, instead, exactly what we observe: a partition of the return distribution into two regimes with different risk-return profiles. An investor who can use this partition to manage exposure (rather than to make on/off trading decisions) captures most of the benefit without incurring most of the cost.

05 Discussion, limitations, and conclusion

Interpretation. The result is a conditioning result, not a prediction result. We do not claim to know whether the next month will be positive or negative. We claim that the distribution from which the next month's return is drawn is materially different depending on whether the index sits above or below its 12-month average. This distinction is important: a forecasting claim would imply substantial alpha; a conditioning claim implies, more modestly, that an investor wishing to manage risk can use the signal to do so cheaply.

Use case: position sizing. The most natural use of the signal is to vary equity exposure modestly based on the regime. A portfolio that holds 100% equity above the MA and 50% equity below would have captured most of the long-run return at materially reduced volatility and drawdown, with simple monthly rebalancing.

Use case: not market timing. A pure on-off strategy (100% equity above MA, 0% below) generates qualitatively similar Sharpe improvements but at substantial cost: tax drag from realising gains, transaction costs from frequent rebalancing, and behavioural costs from holding cash during recovery rallies. We do not recommend this implementation for typical individual investors. Institutional investors with low transaction costs and tax-advantaged structures may find it more attractive.

Limitations. Our analysis is U.S.-only; [Faber \(2007\)](#) documents qualitatively similar results across major developed markets, but emerging-market application is less well-tested. Our zero-risk-free-rate Sharpe is non-standard; the appendix re-computes with a long-rate Sharpe and finds the conditioning result preserved (above-MA Sharpe 0.92, below-MA -0.30). Our drawdown definition is one of several reasonable alternatives. Most importantly, our analysis is on monthly data; daily data would produce somewhat different signals due to the noisier daily price series.

Conclusion. A single binary indicator — above or below the 12-month moving average — partitions 152 years of U.S. equity returns into two materially different conditional distributions. The above-MA distribution carries the bulk of long-run returns at low volatility; the below-MA distribution carries the bulk of long-run drawdown at low return. The result is robust across sub-periods, window lengths, and international markets. It does not constitute alpha in the traditional sense, but it does constitute information — the kind of information that an investor managing risk can use without paying for excessively. The literature has caught up with practitioner intuition on this point; it is a small but worthwhile result.

Carl C.G. Eriksson · The 12-month moving average as a
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*The 12-month MA does not **predict** the next month's return. It **conditions** the distribution from which it is drawn. The distinction is the entire point.*

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APX Appendix A: Robustness with risk-free rate adjustment

Our headline Sharpe calculations use a zero risk-free rate. The historical short-rate series is non-stationary, with periods of near-zero rates (1930s, 2010s) and periods of high rates (1980s). Re-computing using Shiller's long-rate as a proxy for the risk-free rate and assuming the cash leg of any below-MA position earns this rate yields the following:

TABLE A1. SHARPE RATIOS WITH RISK-FREE RATE ADJUSTMENT

Regime	Excess return (ann.)	Sharpe (rf-adj.)
Above MA	+10.85%	0.92
Below MA	-5.04%	-0.30

The qualitative result — large positive Sharpe above MA, small or negative Sharpe below — is preserved under risk-free-rate adjustment. The magnitude of the gap (1.22 in rf-adjusted terms vs 1.26 in raw terms) is essentially identical.

A.2 Position-sizing implementation

A natural use case for the signal is to vary equity exposure between a high allocation (above MA) and a reduced allocation (below MA). The following table shows the historical performance of three example specifications:

A.4 ProRealTime implementation

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// 200-day MA filter, long-only equity-index variant
// ProBacktest / ProOrder, daily timeframe

DEFPARAM CumulateOrders = False
DEFPARAM PreLoadBars = 220

MAlen = 200
ma200 = Average[MAlen](close)

// Long when price > MA, flat otherwise
IF close > ma200 AND NOT LongOnMarket THEN
    BUY 1 CONTRACT AT MARKET
ENDIF

IF close < ma200 AND LongOnMarket THEN
    SELL AT MARKET
ENDIF

// Optional: position size scaled to recent ATR for vol-targeted variant
// atr20 = AverageTrueRange[20](close)
// risk_per_trade = capital * 0.01
// shares = risk_per_trade / (atr20 * 2)
```

TABLE A2. POSITION-SIZED STRATEGIES, 1871–2018

Strategy	Ann. return	Ann. vol	Max DD	Sharpe
100% equity buy-and-hold	+9.31%	14.01%	−81.8%	0.66
100% above / 50% below	+9.16%	10.79%	−61.2%	0.85
100% above / 25% below	+9.18%	9.21%	−54.8%	1.00
100% above / 0% below	+9.24%	9.61%	−49.0%	0.96

All position-sized variants improve risk-adjusted returns relative to buy-and-hold. The improvement is largely from drawdown reduction rather than return enhancement; absolute returns are similar to buy-and-hold across all variants. We note that this analysis ignores transaction costs and taxes; including these would reduce the appeal of the more active variants somewhat but not eliminate it.

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AUTHOR

Carl C.G Eriksson
Founder, ProRealAlgos

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